

A Multi-Task One-Stage Framework for Axial and Directional Object Detection

Is richer instance supervision compatible with higher observed box AP?

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1 Problem & Motivation

Detection answers where and what - not how.

Semantic orientation adds categorical axis and direction labels; it is not rotated-box angle regression.

STANDARD DETECTION

box + class

SEMANTIC ORIENTATION

axis + direction

Axis: O / D
Direction: LR · FR · UD

Research question: Can richer supervision coexist with higher box AP under one shared recipe?

2 Three Contributions

01 Public benchmark

Curated COCO split with box, class, axis, direction and polygon annotations.

02 Controlled study

Four one-stage task heads share a backbone-neck and matched core optimization.

03 Joint head

Separate axis and direction branches join the detection and mask objectives.

Scope transfer: the surrogate transfers the supervision question - not operational X-ray appearance or deployment performance.

3 Public Benchmark

A shared split with richer instance annotations

Every matched object has a box, class, categorical axis, categorical direction and polygon.

IMAGES

5,281

4,753 train
528 validation

MATCHED OBJECTS

31,784

28,659 train
3,125 validation

LABEL STATES

2 + 3

axis + direction

AXIS DISTRIBUTION

Diagonal 97.49%
Orthogonal 2.51%

Severe class imbalance

DIRECTION DISTRIBUTION

Front-Rear 44.85%
Left-Right 44.76%
Up-Down 10.39%

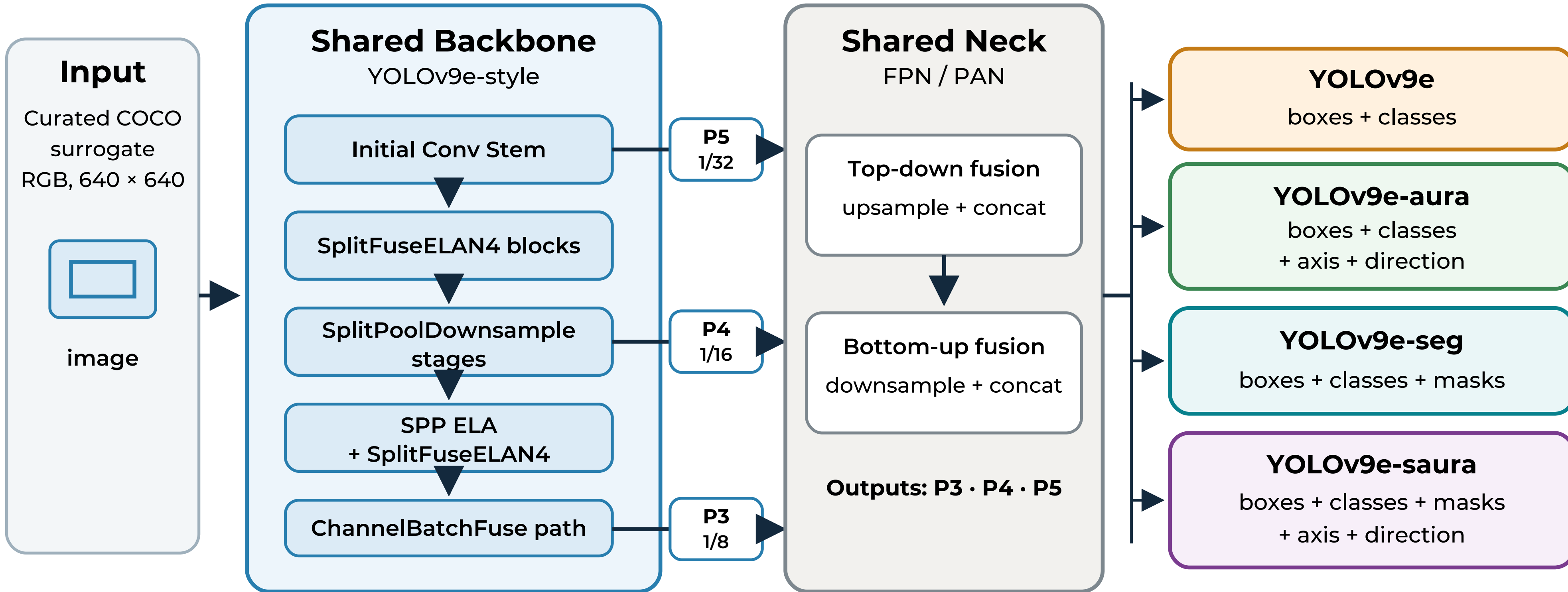


Public COCO image used as a surrogate; not operational X-ray imagery.

Interpretation boundary: the split retains 93 images without a matched object. Polygon completeness applies to matched instances; aggregate axis AP does not prove robustness on the rare Orthogonal state.

4 Method

Simplified architecture overview based on the published Fig. 2: shared backbone-neck, four terminal heads.



Shared features; separate outputs for detector-only, semantic orientation, segmentation, and joint segmentation-orientation heads.

5 Joint Formulation

Separate prediction branches, one weighted objective

$$L = \lambda_{\text{box}} L_{\text{box}} + \lambda_{\text{dfl}} L_{\text{dfl}} + \lambda_{\text{cls}} L_{\text{cls}} + \lambda_{\text{axis}} L_{\text{axis}} + \lambda_{\text{dir}} L_{\text{dir}} + \lambda_{\text{seg}} L_{\text{seg}}$$

- Box loss:** $L_{\text{box}} = 0.45(1 - \text{IoU}) + 0.55(1 - \text{NWD})$.
- Classification, axis and direction use BCE-logits; the mask loss is area-normalized BCE, with Dice off and gains not returned.
- The reported direction head is flat: LR, FR or UD directly, without hierarchical masking.
- Axis and direction do not enter assignment; they only backpropagate.
- Shared features, separate outputs.

6 Experimental Protocol

Matched core optimization, disclosed execution differences

All runs start from scratch on one H100; HSV jitter is active and geometric augmentation is disabled.

EPOCHS

100

INPUT

640 × 640

BATCH

4

OPTIMIZER

SGD

lr 0.01 · momentum 0.937

INITIALIZATION

seed 0

from scratch · AMP

HARDWARE

1 × H100

sequential runs

Matched core: same split, epochs, image size, batch, optimizer family, schedule, seed and precision mode.

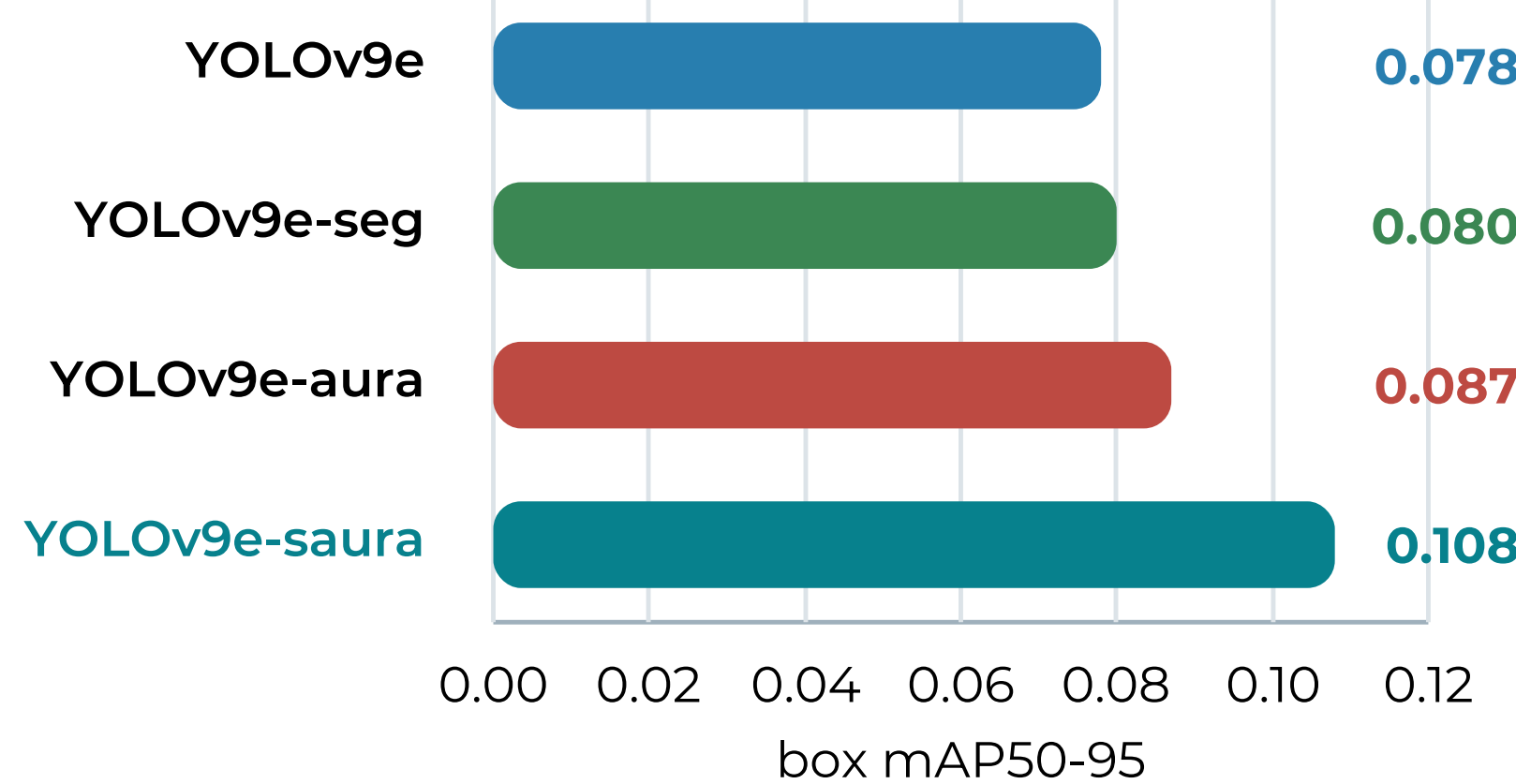
Archived differences: Seg validation cap: 300 detections versus 400. Saura data loader: 4 versus 8 workers, plus prefetch, pinning and persistence differences.

Comparison boundary: published comparisons remain controlled, but capacity is not matched.

7 Main Result

The joint head reaches the highest observed box AP

Best-validation box mAP50-95. Bars are proportional from zero.



+3.0 AP

Saura vs baseline

+0.9 AP

Aura vs baseline

Interpretation: compatible with higher observed AP under this recipe; not causal or statistically significant evidence.

8 Auxiliary Results + Cost

MASK · SEG → SAURA 0.062 → 0.087 +0.025

AXIS · AURA → SAURA 0.124 → 0.215 +0.091

DIRECTION · AURA → SAURA 0.120 → 0.130 +0.010

PARAMETERS 58.15 → 71.08 M +22%

COMPUTE 192.68 → 272.04 GFLOPs +41%

Higher metric-wise peaks come with a larger head; metric-wise peaks may come from different checkpoints.

9 Discussion & Bounded Claim

SUPPORTED HERE

- Richer supervision is compatible with higher observed validation box AP.
- The joint Saura head has the highest displayed metric-wise peaks.
- Public annotations, splits, protocol metadata and reconstruction scripts make the data protocol inspectable.

NOT ESTABLISHED

- Causality or statistical significance: one seed and best validation.
- Compute-matched evidence: capacities and execution details differ.
- Operational X-ray transfer, deployment performance or state of the art.

Correct claim: richer supervision is compatible with higher observed AP in this controlled study on a public surrogate - not proven to cause it.

+3.0 AP

Saura versus baseline
box mAP50-95



PUBLIC REPOSITORY

[github.com/mbenhamd/
avss2026-curated-coco-benchmark](https://github.com/mbenhamd/avss2026-curated-coco-benchmark)

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